LUMINAR

E-COMMERCE SALES & CUSTOMER ANALYTICS

SQL PROJECT REPORT

Submitted as part of the Data Analytics Course

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Abstract

This project uses MySQL to organize and analyze data for a simulated online store. The database includes customers, sellers, products, orders, order items, and payments. SQL queries address customer activity, order status, product performance, revenue, payment methods, and repeat purchasing. Advanced techniques include joins, aggregates, subqueries, CTEs, window functions, a view, and a stored procedure.

1. Introduction

E-commerce data can help businesses understand product performance, customer spending, sales trends, and payment-method usage. This project is part of the Data Analytics course at Luminar and uses MySQL Workbench.

2. Problem Statement

The project builds a relational database for an online store and uses SQL queries to answer business questions through filtering, sorting, aggregation, joins, and advanced techniques.

3. Objectives

• Create and use a MySQL database.

• Organize data into six related tables.

• Use filtering, sorting, joins, aggregate functions, subqueries, CTEs, and window functions.

• Create a monthly sales view and customer-history procedure.

• Document design, queries, and verified outputs.

4. Tools and Technologies

• MySQL — relational database management system.

• MySQL Workbench — SQL execution and EER diagram creation.

• SQL — data retrieval and analysis.

5. Database Design

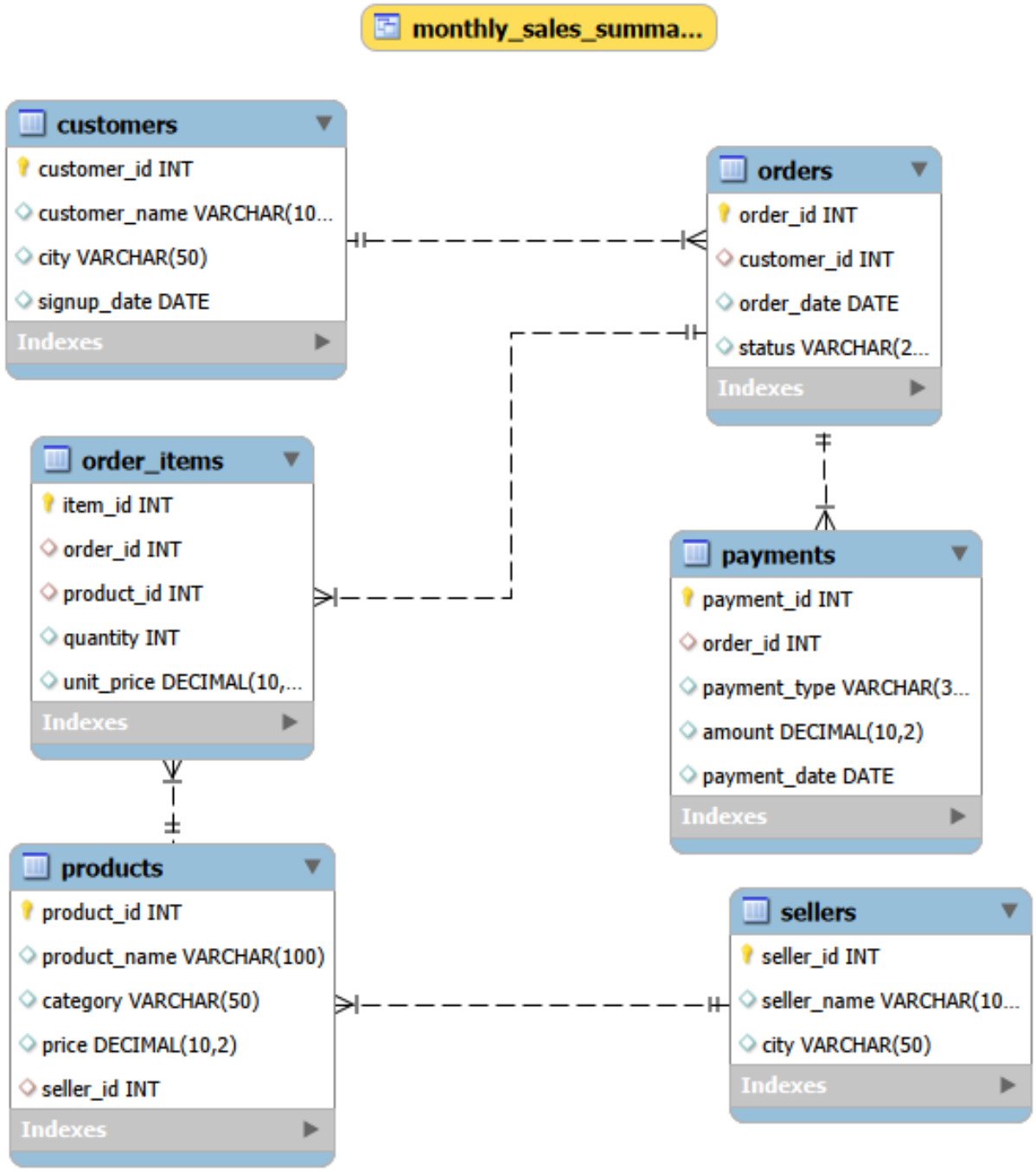
The ecommerce_project schema contains six main tables.

Table	Purpose	Analytical use
customers	Customer details	City/date filters and spending
sellers	Seller details	Seller sales comparison
products	Product information	Price/category/revenue
orders	Order details	Dates/status/customer activity
order_items	Products and quantities per order	Units sold/revenue
payments	Payment records	Revenue/payment types

Primary keys identify records, and relationship lines connect related tables.

Figure 1 — Entity Relationship Diagram

The diagram shows the six main tables and monthly_sales_summary view.



6. Methodology

Database setup

Created and selected ecommerce_project in MySQL Workbench.

Table creation and data

Created six tables and used available sample records.

Basic queries

Filtered and sorted customer, order, and product records.

Intermediate analysis

Used joins and aggregates for revenue, units sold, customer spending, seller performance, order value, and payment types.

Advanced analysis

Used subqueries, CTEs, window functions, a view, and a stored procedure.

Validation

Executed queries and reviewed outputs in MySQL Workbench.

7. Business Questions and SQL Concepts

• Beginner: customers by city, recent orders, expensive products, delivered orders, 2025 signups — SELECT, WHERE, ORDER BY, LIMIT.

• Intermediate: revenue, highest-selling category, top customers, seller threshold, order value by city, payment summaries — JOIN, GROUP BY, HAVING, SUM, COUNT, AVG.

• Advanced: product rankings, above-average spend, repeat customers, monthly sales view, customer history — subqueries, CTEs, window functions, views, procedures.

8. Advanced SQL Features

Window functions

RANK() or DENSE_RANK() ranks products by revenue within each category.

CTEs

A CTE introduced with WITH defines a named result set for multi-step analysis.

monthly_sales_summary view

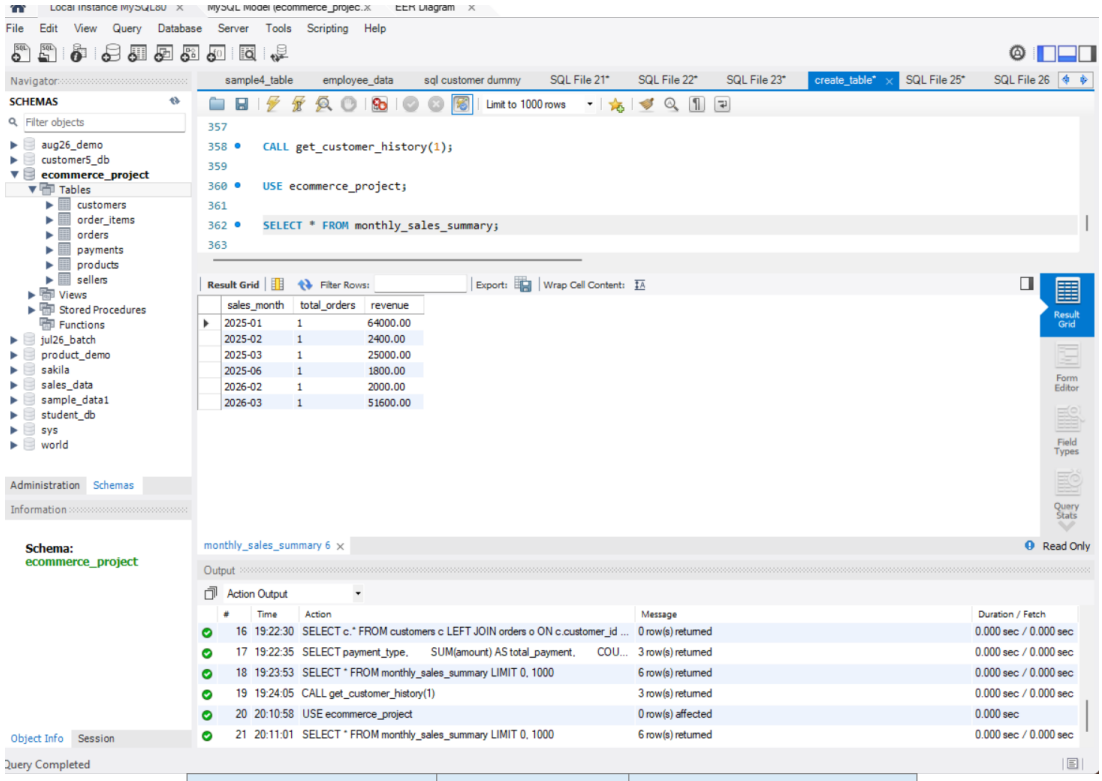
This view provides a reusable monthly sales query.

get_customer_history stored procedure

The procedure accepts a customer ID and returns that customer's order history.

9. Verified Results — Monthly Sales Summary

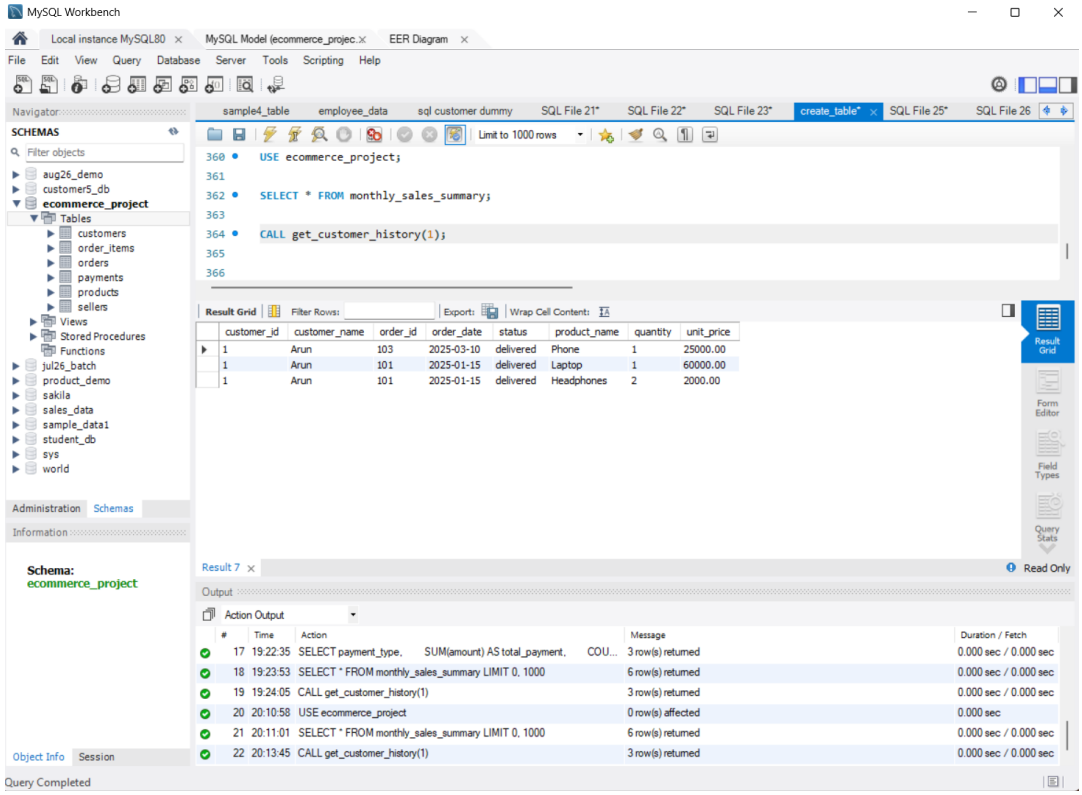
The query executed successfully and returned six rows.



Month	Orders	Revenue
2025-01	1	64,000.00
2025-02	1	2,400.00
2025-03	1	25,000.00
2025-06	1	1,800.00
2026-02	1	2,000.00
2026-03	1	51,600.00

10. Verified Results — Customer History

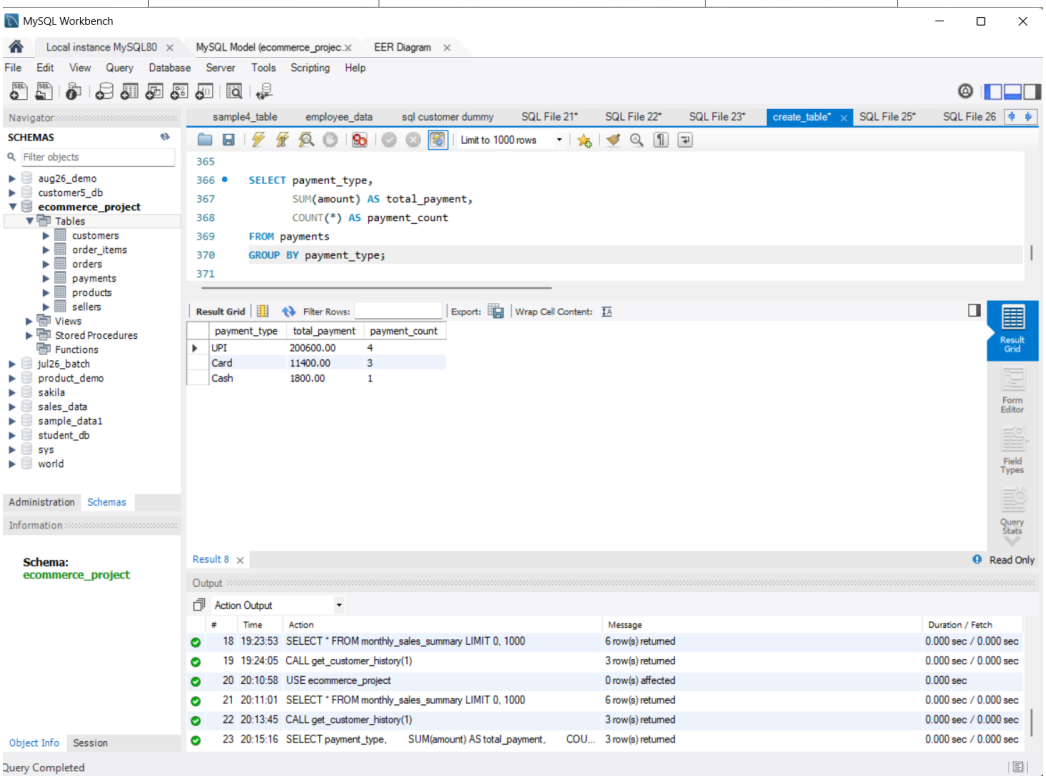
CALL get_customer_history(1); executed successfully and returned three rows for customer Arun, with order-item details for Phone, Laptop, and Headphones.



11. Verified Results — Payment Type

The payment summary query executed successfully and returned three payment types:

Type	Total payment	Count
UPI	200,600.00	4
Card	11,400.00	3
Cash	1,800.00	1



12. Key Findings

• Six months are shown in the monthly view; each displayed month has one order.

• Highest displayed monthly revenue: 64,000.00 in January 2025; lowest: 1,800.00 in June 2025.

• UPI has the highest total payment amount in the displayed payment summary: 200,600.00 across four records.

• The customer-history procedure returned three rows for customer ID 1 (Arun).

These findings describe only the displayed query outputs, not necessarily complete business totals.

13. Skills Practiced

Relational database design, filtering, sorting, joins, aggregates, subqueries, CTEs, window functions, views, stored procedures, and query validation.

14. Conclusion

The project demonstrates how SQL can organize related e-commerce data and answer practical business questions. The database, queries, EER diagram, and verified screenshots can form part of a Data Analytics portfolio.

15. Future Enhancements

• Connect the monthly sales view to Power BI or Tableau.

• Add records and validate data quality.

• Explore sales growth, category trends, and customer segmentation.

• Publish SQL scripts, the EER diagram, and screenshots in GitHub.

Appendix — Submission Checklist

• SQL table-creation script.

• Beginner, intermediate, and advanced queries.

• EER diagram.

• Monthly sales view screenshot.

• Customer-history procedure screenshot.

• Payment-type summary screenshot.

• Additional verified query screenshots.

• README.md.

Credit any external dataset used; if sample data was created manually, describe it as sample/custom-built data.